



ADIKAVI NANNAYA UNIVERSITY::RAJAMAHENDRAVARAM
UNIVERSITY COLLEGE OF ENGINEERING

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

Academic Year: 2025-2026

Course: III BTech, V Semester

Name of the Exam: MID-II

Date: -10-2025

Duration: 90 Min.

Max. Marks: 40

SUBJECT: EMWTL

SECTION-A

Answer the following Questions

(3x10=30)

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1. A.[i] Explain Cartesian, Cylindrical and Spherical coordinate systems with neat diagrams.
[ii] State and explain Coulomb's Law. Derive the expression for Electric Field Intensity.

(OR)

B.[i] State and explain Gauss Law in differential form and also list the limitations and applications of Gauss Law.

- [ii] A circular ring of charge with radius 5m lied in $Z=0$ plane with centre at origin. If the line charge density is 10 nC/m . Find the point charged a placed at the origin. which will produce the same E at point (0, 6)m.

2. A) [i] Derive Poisson's and Laplace's equations and explain their physical significance.
[ii] What is meant by Electric Field Intensity?

Electric Field Intensity due to:

- i)Infinite line charge ii)Infinite sheet charge iii)Infinite volume charge

(OR)

B) Explain Capacitance and derive expressions for:

- i)Parallel plate capacitor ii)Co-axial capacitor

3. A) [i] State and explain Biot-Savart Law with applications.
[ii] A circular loop of wire of radius 'r' lying in xy-plane with its centre at origin carries a current 'I' is the $+\Phi$ direction. Using Biot-Savart Law, find $H(0,0,Z)$ and $H(0,0,0)$.

(OR)

B) [i] State Ampere's Circuital Law. Derive the expression for the point form of Ampere's Circuital Law.

- [ii] Write the Maxwell's equations for harmonically varying fields.

SECTION- B (5×2=10M)

Answer all the following Questions

- 4) Define gradient, divergence, and curl.
- 5) Write the expression for electric flux density.
- 6) State Faraday's Law.
- 7) Write applications of Ampere's Law.
- 8) Define co-efficient of coupling.